

Local Non-Bossiness and Improvements over Deferred Acceptance*

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Abstract

In school choice, the student-proposing deferred acceptance mechanism (DA) is stable and strategy-proof, but it may be Pareto inefficient for students. Duque-Rosas, Pereyra, and Torres-Martínez (2026) show that DA is locally non-bossy: a student who changes her report and keeps her school cannot change who else is assigned to that school. We ask whether mechanisms that Pareto improve upon DA inherit this protection. Any improvement over DA preserves the size of every school and the set of unassigned students; hence, when schools have a single seat, every mechanism that improves upon DA is locally non-bossy. With multi-seat schools the conclusion fails: we exhibit an environment in which no Pareto-efficient mechanism that improves upon DA is locally non-bossy. Efficiency adjustments can therefore give a student leverage over her classmates even when they give her none over her school.

Keywords: school choice; deferred acceptance; Pareto efficiency; local non-bossiness.

JEL classification: C78, D47, D61, I20.

1 Introduction

School choice systems assign seats using two inputs: the preferences that students report and the priorities that schools hold over them, which typically encode administrative or legal claims rather than the schools' own objectives. Two requirements organize the design problem. *Stability* asks that no student prefer a school that either leaves a seat vacant or admits a student with lower priority, and *strategy-proofness* asks that no family gain by misreporting preferences. The student-proposing deferred acceptance mechanism (DA) satisfies both and selects the stable matching that every student weakly prefers among the stable ones (Dubins & Freedman, 1981; Gale & Shapley, 1962; Roth, 1982). This is why DA is the benchmark of the literature and of practice.

The benchmark has a familiar cost. DA need not be *Pareto efficient for students*: there may exist another feasible assignment that makes every student weakly better off and at least one student strictly better off. This observation motivates mechanisms that start from the DA outcome and recover some or all of the foregone gains. The leading example is the *efficiency-adjusted deferred acceptance* mechanism (EADA) of Kesten (2010). A student whose own

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assignment does not depend on a given priority may *consent* to waive it, which allows others to trade seats without harming her; when every student consents, an assumption known as *full consent*, EADA produces a Pareto-efficient matching that weakly Pareto dominates DA. Tang and Yu (2014) give an equivalent formulation of full-consent EADA that dispenses with consent decisions: repeatedly settle the students assigned to *underdemanded* schools, that is, schools that no student prefers to her current assignment, and rerun DA in the remaining market.

Improving DA changes other desirable features of the benchmark. An efficient improvement over an inefficient DA outcome cannot remain stable, because DA is the best stable matching for students. More generally, no strategy-proof mechanism can Pareto improve upon DA on the unrestricted domain (Abdulkadiroğlu, Pathak, & Roth, 2009; Alva & Manjunath, 2019; Kesten, 2010). The literature has therefore studied weaker disciplines on such improvements. For example, Doğan and Yenmez (2020) study *consistency*, the requirement that removing assigned students together with the seats they occupy should not alter the assignments of the remaining students, and show that no consistent Pareto improvement over a stable mechanism exists.¹ Doğan and Ehlers (2021) and Knipe and Ortega (2025) study how much instability must accompany welfare improvements, comparing assignments by their blocking pairs and by the number of students who can be improved. Ortega et al. (2026) identify rank and distributional shortcomings that Pareto-efficiency adjustments cannot repair.

This note studies a different property of DA. A mechanism is *non-bossy* if a student who changes her report without changing her own assignment cannot change anybody else’s assignment. DA does not satisfy this global property: a student’s report can change assignments elsewhere in the market while leaving her own school unchanged. Duque-Rosas, Pereyra, and Torres-Martínez (2026) show, however, that DA’s bossiness has a sharp local limit. If the student keeps the same school, she cannot change the set of students assigned to that school. They call this property *local non-bossiness*. It matters when students care about their classmates: ordinary strategy-proofness prevents a profitable lie that changes one’s school, whereas local non-bossiness prevents a lie that keeps one’s school but changes one’s classmates. Duque-Rosas, Pereyra, and Torres-Martínez (2026) use this observation to recover strategy-proofness when students first rank their own school and, conditional on that school, care only about their schoolmates.

We ask whether this local protection survives Pareto improvements over DA. The answer splits sharply with capacities. Our first result is an invariance property. If an individually rational and non-wasteful matching is weakly Pareto improved, then the number of students assigned to every school and the set of unassigned students remain unchanged. Applied to DA, this means that a DA-dominating mechanism only reshuffles occupied seats. It cannot fill a previously empty seat or assign a student whom DA leaves unassigned. This immediately implies that a DA-dominating mechanism can never violate local non-bossiness through the outside option. Moreover, if every school has one seat, a student who keeps her school necessarily keeps its entire composition. Hence every DA-dominating mechanism is locally non-bossy in one-to-one problems.

The result changes once a school can admit more than one student. We construct a five-student, four-school environment with one two-seat school and two preference profiles that differ only in one student’s report. At each profile there is exactly one strict Pareto improvement over DA. Any mechanism that is both Pareto efficient and DA-dominating is therefore forced to select

¹Formally, consistency compares the mechanism’s outcome in an original problem with its outcome in a reduced problem obtained by removing a set of students and reducing each school’s capacity by the number of removed students assigned there. The remaining students must keep their original assignments. See Ergin (2002) and Doğan and Yenmez (2020).

that improvement. The same student is assigned to the two-seat school under both reports, but the identity of her classmate changes. Thus no efficient DA-dominating mechanism can be locally non-bossy in that environment. Efficiency is stronger than the proof needs: the contradiction only requires the mechanism to implement the available strict improvement at the two profiles.

Two implications make the distinction economically relevant. First, full-consent EADA is locally bossy in our environment. Second, on the domain of school-lexicographic preferences over colleagues studied by Duque-Rosas, Pereyra, and Torres-Martínez (2026), every efficient DA-dominating mechanism is manipulable. The manipulating student cannot obtain a better school: under both reports every mechanism in the class assigns her to the same school. Her report instead determines which student becomes her classmate. Efficiency adjustments therefore create a strategic channel that DA closes, and they do so precisely when the student’s own school assignment does not move.

Related literature. The efficiency cost of priority-respecting assignments is studied by Balinski and Sönmez (1999) and Abdulkadiroğlu and Sönmez (2003). Ergin (2002) characterizes acyclic priority structures, under which DA is Pareto efficient, *group strategy-proof*, and consistent.² Efficient improvements over DA are studied by Kesten (2010), Tang and Yu (2014), Tang and Zhang (2021), and Troyan, Delacrétaz, and Kloosterman (2020). Their stability and the extent of their improvements are studied, among others, by Doğan and Ehlers (2021), Afacan et al. (2026), and Knipe and Ortega (2025). On incentives, Abdulkadiroğlu, Pathak, and Roth (2009) show that no strategy-proof mechanism Pareto improves upon DA, Kesten and Kurino (2019) identify maximal domains for such improvements, Alva and Manjunath (2019) characterize strategy-proof Pareto improvements, and Cerrone, Hermstrüwer, and Kesten (2024) study consent and truth-telling experimentally. Non-bossiness goes back to Satterthwaite and Sonnenschein (1981); see Thomson (2016) for a survey, Pápai (2000) for its relation with group strategy-proofness, and Kojima (2010) for the incompatibility between stability and non-bossiness when both sides report.

The paper closest to this note is Duque-Rosas, Pereyra, and Torres-Martínez (2026). They introduce local non-bossiness, prove that DA satisfies it, and show how the property governs incentives when students care about their schoolmates. They also discuss EADA to distinguish the improvement cycles used in their proof from those implemented by Kesten’s mechanism, and explicitly note that their results are not directly about EADA. They do not ask whether Pareto improvements over DA preserve local non-bossiness. Our results address that question and identify a capacity boundary: DA-dominance is automatically locally non-bossy with unit capacities, while efficient DA-dominating mechanisms can be forced to violate the property once a school has multiple seats.

The rest of the note is organized as follows. Section 2 presents the model. Section 3 establishes the invariance property and the one-to-one possibility result. Section 4 contains the many-to-one impossibility. Section 5 derives the implications for EADA and for preferences over classmates, and Section 6 concludes.

²A mechanism is group strategy-proof if no coalition can jointly misreport so that every member is weakly better off and at least one member is strictly better off.

2 Model

There is a finite set of students I and a finite set of schools C . Each school $c \in C$ has capacity $q_c \in \mathbb{N}$ and a strict priority order $\succ_c$ over I . Each student $i \in I$ has a strict preference relation P_i over $C \cup \{\emptyset\}$, where $\emptyset$ is the outside option; R_i denotes the associated weak relation. A school c is *acceptable* to i if $c P_i \emptyset$. We write $P = (P_i)_{i \in I}$ and let $\mathcal{P}$ be the unrestricted domain of such profiles. The environment $\Gamma = (I, C, q, \succ)$ is fixed throughout, and only P varies.

An assignment $\mu : I \rightarrow C \cup \{\emptyset\}$ is *feasible* if no school exceeds its capacity. We use the term *matching* for a feasible assignment. For every $x \in C \cup \{\emptyset\}$, let

$$\mu_x = \{i \in I : \mu_i = x\}.$$

Thus $|\mu_c| \leq q_c$ for every $c \in C$ and $\mu_\emptyset$ is the set of unassigned students. Let $\mathcal{M}$ denote the set of matchings; since I and C are finite, so is $\mathcal{M}$. A matching is *individually rational* if $\mu_i R_i \emptyset$ for every i , and *non-wasteful* if there is no student-school pair (i, c) such that i prefers c to μ_i while c has an empty seat.

A matching μ is *stable* if it is individually rational and there is no pair (i, c) such that $c P_i \mu_i$ and either $|\mu_c| < q_c$ or $i \succ_c j$ for some $j \in \mu_c$. Such a pair is a *blocking pair*: student i prefers c to her assignment and can claim either an empty seat or a seat held by a lower-priority student. Thus stability is equivalent here to individual rationality, non-wastefulness, and absence of justified envy.³

Student-proposing deferred acceptance (DA).

1. Each student applies to her most-preferred acceptable school. Each school tentatively keeps up to q_c highest-priority applicants and rejects the rest.
2. Every rejected student applies to her most-preferred acceptable school that has not yet rejected her. Each school again keeps the q_c highest-priority students among its current tentative assignees and new applicants, rejecting the rest.
3. Repeat Step 2 until no rejection occurs. Tentative assignments then become final.

The resulting matching is denoted $DA(P)$. It is the student-optimal stable matching: every student weakly prefers it to every other stable matching at P (Gale & Shapley, 1962).

Given P , a matching ν *weakly Pareto dominates* μ if $\nu_i R_i \mu_i$ for every $i \in I$. It *Pareto dominates* μ if, in addition, $\nu_j P_j \mu_j$ for some j . A matching is *Pareto efficient* if no matching Pareto dominates it.

A *mechanism* is a map $\varphi : \mathcal{P} \rightarrow \mathcal{M}$. We write $\varphi_i(P)$ for student i 's assignment and $\varphi_x(P) = \{i \in I : \varphi_i(P) = x\}$ for $x \in C \cup \{\emptyset\}$. A mechanism is *strategy-proof* if no student can obtain a strictly preferred assignment by changing only her own report: there are no $i \in I$, $P \in \mathcal{P}$, and P'_i such that $\varphi_i(P'_i, P_{-i}) P_i \varphi_i(P)$. DA is strategy-proof for students (Dubins & Freedman, 1981; Roth, 1982).

Definition 1. A mechanism φ is *DA-dominating* if $\varphi(P)$ weakly Pareto dominates $DA(P)$ for every $P \in \mathcal{P}$. It is *efficient* if $\varphi(P)$ is Pareto efficient for every $P \in \mathcal{P}$.

³A student has *justified envy* at a matching when she prefers some school to her own assignment and that school admits a student with lower priority for it. Priorities are respected in this weak sense: a student may envy a classmate of a school she cannot claim, but never one who ranks below her.

Definition 2 (Duque-Rosas, Pereyra, and Torres-Martínez (2026)). A mechanism φ is *locally non-bossy* if for every $i \in I$, every $P \in \mathcal{P}$, every report P'_i , and every $x \in C \cup \{\emptyset\}$,

$$\varphi_i(P) = \varphi_i(P'_i, P_{-i}) = x \implies \varphi_x(P) = \varphi_x(P'_i, P_{-i}).$$

In words, a student who reports different preferences and is still assigned to the same school cannot change the composition of that school, and a student who reports different preferences and remains unassigned cannot change the set of unassigned students. The latter condition makes local non-bossiness non-vacuous even in one-to-one problems (Duque-Rosas, Pereyra, & Torres-Martínez, 2026). Local non-bossiness is weaker than *non-bossiness*: a non-bossy mechanism requires the entire matching to remain unchanged whenever the reporting student keeps her assignment.⁴

Proposition 1 (Duque-Rosas, Pereyra, and Torres-Martínez (2026), Theorem 1). *The student-optimal stable mechanism is locally non-bossy.*

3 Improvements over DA preserve school sizes

We first establish a property of Pareto improvements that does not depend on DA itself. The observation is closely related to the argument used by Abdulkadiroğlu, Pathak, and Roth (2009) to show that a Pareto improvement over a stable matching preserves the set of matched students.

Lemma 1. *Let μ be individually rational and non-wasteful at P , and let ν weakly Pareto dominate μ at P . Then*

$$|\nu_c| = |\mu_c| \quad \text{for every } c \in C, \quad \text{and} \quad \nu_\emptyset = \mu_\emptyset.$$

Proof. If $\mu_i \in C$, individual rationality and strict preferences give $\mu_i P_i \emptyset$. Weak Pareto dominance therefore implies $\nu_i \neq \emptyset$, so $\nu_\emptyset \subseteq \mu_\emptyset$.

Suppose $|\nu_c| > |\mu_c|$ for some $c \in C$. Feasibility of ν implies $|\mu_c| < q_c$, and there exists $i \in \nu_c \setminus \mu_c$. Since $\nu_i = c \neq \mu_i$ and preferences are strict, weak Pareto dominance gives $c P_i \mu_i$. Thus i prefers a school that has a vacant seat under μ , contradicting non-wastefulness. Hence $|\nu_c| \leq |\mu_c|$ for every school.

Summing these inequalities gives $\sum_c |\nu_c| \leq \sum_c |\mu_c|$. But $\nu_\emptyset \subseteq \mu_\emptyset$ gives the reverse inequality for the numbers of assigned students. Therefore equality holds at every school and $|\nu_\emptyset| = |\mu_\emptyset|$. Together with $\nu_\emptyset \subseteq \mu_\emptyset$, this yields $\nu_\emptyset = \mu_\emptyset$. $\square$

Let φ be a DA-dominating mechanism and fix a profile P . The matching $\text{DA}(P)$ is stable, hence individually rational and non-wasteful, and $\varphi(P)$ weakly Pareto dominates it. Lemma 1, applied to $\mu = \text{DA}(P)$ and $\nu = \varphi(P)$, therefore gives

$$|\varphi_c(P)| = |\text{DA}_c(P)| \quad \text{for all } c \in C, \quad \varphi_\emptyset(P) = \text{DA}_\emptyset(P).$$

Thus improvements over DA can reshuffle occupied seats but cannot change school sizes or rescue a student whom DA leaves unassigned.

Proposition 2. *Let φ be DA-dominating. Fix $P \in \mathcal{P}$, $i \in I$, and another report P'_i , and let $P' = (P'_i, P_{-i})$.*

⁴Formally, φ is non-bossy if $\varphi_i(P) = \varphi_i(P'_i, P_{-i})$ implies $\varphi(P) = \varphi(P'_i, P_{-i})$ (Satterthwaite & Sonnenschein, 1981). Local non-bossiness places no restriction on what the report of i does at schools other than $\varphi_i(P)$.

(i) If $\varphi_i(P) = \varphi_i(P') = \emptyset$, then $\varphi_\emptyset(P) = \varphi_\emptyset(P')$.

(ii) If $\varphi_i(P) = \varphi_i(P') = c \in C$ and $\varphi_c(P) \neq \varphi_c(P')$, then $q_c \geq 2$.

Proof. For part (i), suppose $\varphi_i(P) = \emptyset$. Because $\varphi(P)$ weakly Pareto dominates the individually rational matching $DA(P)$, student i must also be unassigned under $DA(P)$; otherwise $DA_i(P) P_i \emptyset = \varphi_i(P)$. The same argument gives $DA_i(P') = \emptyset$. Proposition 1, applied to the outside option, therefore gives $DA_\emptyset(P) = DA_\emptyset(P')$. Lemma 1 gives $\varphi_\emptyset(P) = DA_\emptyset(P)$ and $\varphi_\emptyset(P') = DA_\emptyset(P')$, proving the claim.

For part (ii), student i belongs to both $\varphi_c(P)$ and $\varphi_c(P')$. If $q_c = 1$, feasibility forces both sets to equal $\{i\}$, contradicting $\varphi_c(P) \neq \varphi_c(P')$. Hence $q_c \geq 2$. $\square$

Corollary 1. *If $q_c = 1$ for every $c \in C$, then every DA-dominating mechanism is locally non-bossy. In particular, in every one-to-one environment there exists a mechanism that is efficient, DA-dominating, and locally non-bossy.*

Proof. At the outside option the claim follows from Proposition 2(i). At a proper school it follows from Proposition 2(ii), because unit capacity rules out a change in the school's composition when the reporting student remains assigned there.

For existence, fix a profile P and consider the set of matchings that weakly Pareto dominate $DA(P)$. It is non-empty, since $DA(P)$ belongs to it, and it is finite because $\mathcal{M}$ is finite. Choose a maximal element of this set under Pareto dominance.⁵ Such an element is Pareto efficient: if some matching Pareto dominated it, transitivity would imply that this matching also weakly Pareto dominates $DA(P)$, contradicting maximality. Making such a selection at every profile defines an efficient and DA-dominating mechanism, which is locally non-bossy by the first part of the corollary. $\square$

Local non-bossiness is therefore compatible with improving DA whenever every school has one seat, and the compatibility is automatic rather than the result of a careful choice of mechanism. We next show that this conclusion does not extend to many-to-one environments.

4 A many-to-one impossibility

4.1 The environment

Let $\Gamma^0 = (I, C, q, \succ)$ be the environment with

$$I = \{i_1, i_2, i_3, i_4, i_5\}, \quad C = \{c_1, c_2, c_3, c_4\},$$

capacities

$$q_{c_1} = 2, \quad q_{c_2} = q_{c_3} = q_{c_4} = 1,$$

and the priority profile displayed in Table 1. Total capacity therefore equals the number of students. The table also displays the two preference profiles used below, $P = (P_{i_1}, P_{-i_1})$ and $P' = (P'_{i_1}, P_{-i_1})$. All schools are acceptable to all students, so the outside option plays no role in the example.

⁵An element is *maximal* in a set under Pareto dominance if no other element of the set Pareto dominates it. A maximal element exists in any finite non-empty set, since Pareto dominance is a strict partial order.

Priorities	Preferences
$\succ_{c_1}: i_2, i_3, i_1, i_4, i_5$	$P_{i_1}: c_1, c_2, c_3, c_4$
$\succ_{c_2}: i_1, i_4, i_2, i_3, i_5$	$P'_{i_1}: c_1, c_3, c_2, c_4$
$\succ_{c_3}: i_1, i_5, i_3, i_2, i_4$	$P_{i_2}: c_2, c_1, c_3, c_4$
$\succ_{c_4}: i_4, i_5, i_1, i_2, i_3$	$P_{i_3}: c_3, c_1, c_2, c_4$
	$P_{i_4}: c_2, c_4, c_1, c_3$
	$P_{i_5}: c_3, c_4, c_1, c_2$

Table 1: The environment Γ^0 and the two preference profiles.

The two profiles differ only in the report of i_1 , and only in the order of c_2 and c_3 , the two schools immediately below her top choice. Under either report, student i_1 is eventually rejected by c_1 ; her report determines where she applies after that rejection. Running the algorithm gives

$$\text{DA}(P) : \quad c_1 = \{i_2, i_3\}, \quad c_2 = \{i_1\}, \quad c_3 = \{i_5\}, \quad c_4 = \{i_4\},$$

$$\text{DA}(P') : \quad c_1 = \{i_2, i_3\}, \quad c_2 = \{i_4\}, \quad c_3 = \{i_1\}, \quad c_4 = \{i_5\}.$$

The computation is immediate. At P , school c_2 holds i_4 and rejects i_2 , school c_3 holds i_5 and rejects i_3 , and c_1 then receives i_1, i_2 and i_3 and keeps its two highest-priority applicants, i_2 and i_3 . Student i_1 next applies to c_2 , displaces i_4 because $i_1 \succ_{c_2} i_4$, and i_4 is accepted at c_4 . At P' the algorithm is identical until i_1 is rejected by c_1 ; she then applies to c_3 , displaces i_5 because $i_1 \succ_{c_3} i_5$, and i_5 is accepted at c_4 , while i_4 keeps c_2 .

4.2 Only one way to improve

For a profile P , let $\mathcal{D}(P)$ denote the set of matchings that weakly Pareto dominate $\text{DA}(P)$ at that profile.

Lemma 2. *In Γ^0 ,*

$$\mathcal{D}(P) = \{\text{DA}(P), \nu\}, \quad \nu : \quad c_1 = \{i_1, i_3\}, \quad c_2 = \{i_2\}, \quad c_3 = \{i_5\}, \quad c_4 = \{i_4\},$$

and

$$\mathcal{D}(P') = \{\text{DA}(P'), \nu'\}, \quad \nu' : \quad c_1 = \{i_1, i_2\}, \quad c_2 = \{i_4\}, \quad c_3 = \{i_3\}, \quad c_4 = \{i_5\}.$$

Moreover, ν is Pareto efficient at P and ν' is Pareto efficient at P' .

Proof. Consider P and write $\mu = \text{DA}(P)$. Student i_5 is assigned to her top choice c_3 , so she remains at c_3 in every element of $\mathcal{D}(P)$. Student i_3 is assigned to c_1 under μ and only prefers c_3 , which is fixed for i_5 ; hence i_3 remains at c_1 . Student i_1 can weakly improve upon c_2 only by moving to c_1 or remaining at c_2 ; student i_2 can weakly improve upon c_1 only by moving to c_2 or remaining at c_1 ; and student i_4 can weakly improve upon c_4 only by moving to c_2 or remaining at c_4 .

Since i_3 occupies one of the two seats at c_1 , the remaining seat is occupied by either i_1 or i_2 . If it is occupied by i_1 , then i_2 must be assigned to c_2 and i_4 to c_4 , yielding ν . If it is occupied by i_2 , then i_1 remains at c_2 and i_4 at c_4 , yielding μ . Student i_4 cannot instead occupy c_2 : that

would force i_1 to c_1 , leaving i_2 without an assignment weakly preferred to her DA assignment c_1 . Thus $\mathcal{D}(P) = \{\mu, \nu\}$.

If a matching Pareto dominated ν , transitivity would imply that it weakly Pareto dominates μ , so it would belong to $\mathcal{D}(P)$. Since neither element of $\mathcal{D}(P)$ Pareto dominates ν , ν is Pareto efficient.

At P' the argument is symmetric. Student i_4 is at her top choice c_2 and is fixed there. This prevents i_2 from improving from c_1 to c_2 , so i_2 is fixed at c_1 . Student i_1 can be assigned to c_1 or c_3 , student i_3 to c_3 or c_1 , and student i_5 to c_3 or c_4 . The free seat at c_1 is occupied by i_1 , yielding ν' , or by i_3 , yielding $\text{DA}(P')$. Student i_5 cannot occupy c_3 in the former case, because then i_3 would have no assignment weakly preferred to her DA assignment c_1 . Therefore $\mathcal{D}(P') = \{\text{DA}(P'), \nu'\}$, and the same argument proves the efficiency of ν' . $\square$

Dominance is evaluated at the reported profile. In particular, at P' the report P'_{i_1} is used when evaluating whether a matching weakly Pareto dominates $\text{DA}(P')$. Local non-bossiness is likewise a property of reports, so no separate notion of “true” preferences is needed for the result below.

4.3 The impossibility

Theorem 1. *In Γ^0 , no mechanism is simultaneously efficient, DA-dominating, and locally non-bossy.*

Proof. Let φ be efficient and DA-dominating. By Lemma 2, $\mathcal{D}(P) = \{\text{DA}(P), \nu\}$ and ν Pareto dominates $\text{DA}(P)$. Efficiency therefore forces $\varphi(P) = \nu$. The same argument at P' gives $\varphi(P') = \nu'$. Hence

$$\varphi_{i_1}(P) = c_1 = \varphi_{i_1}(P'),$$

while

$$\varphi_{c_1}(P) = \{i_1, i_3\} \neq \{i_1, i_2\} = \varphi_{c_1}(P').$$

Since P and P' differ only in the report of i_1 , local non-bossiness fails. $\square$

Efficiency is stronger than the argument needs. At the two profiles of Γ^0 the proof only requires the mechanism to weakly Pareto dominate DA and to select a strict improvement rather than reproduce DA. The conflict therefore arises already for mechanisms that are required to implement an available improvement at these two profiles, even if they are not efficient elsewhere.

Remark 1 (Outside options and capacity). The example is balanced and every school is acceptable to every student, so neither unmatched students nor outside options drive the impossibility. One can delete c_4 and place the outside option immediately below c_2 for i_4 and below c_3 for i_5 ; the same argument then works with three schools and one student unassigned at each of the two DA outcomes. By Proposition 2, however, a school with at least two seats is indispensable for any violation by a DA-dominating mechanism.

Remark 2 (Relation to the strategy-proofness impossibility). Abdulkadiroğlu, Pathak, and Roth (2009) show that no strategy-proof mechanism Pareto improves upon DA on the unrestricted domain. Their restriction concerns whether a student can improve her own assignment by changing her report. Local non-bossiness asks a different question: can she keep her assignment but change who is assigned with her? Neither statement implies the other, and Theorem 1 does not use strategy-proofness. In our example, every efficient DA-dominating mechanism assigns i_1 to c_1 under both reports; only the composition of c_1 changes.

5 Implications

5.1 Efficiency-adjusted deferred acceptance

For completeness, we use the simplified formulation of full-consent EADA due to Tang and Yu (2014), which produces the same matching as Kesten’s mechanism when every student consents to waive the priorities that do not determine her own assignment.

Full-consent efficiency-adjusted deferred acceptance (EADA).

1. Run DA in the current market.
2. A school is *underdemanded* at the resulting matching if no student strictly prefers that school to her current assignment. Permanently settle every underdemanded school together with the students assigned to it, and remove those schools and students from the market.
3. If students remain, rerun DA in the reduced market and repeat Step 2. Stop once all students and schools are settled.

This algorithm is well defined, weakly Pareto dominates DA, and is Pareto efficient (Kesten, 2010; Tang & Yu, 2014).

Corollary 2. *In Γ^0 , full-consent EADA is locally bossy.*

Proof. Full-consent EADA is efficient and DA-dominating, so Theorem 1 applies. The outcome can also be seen directly. At P , the only underdemanded school at $DA(P)$ is c_4 . Settling i_4 at c_4 and rerunning DA among $\{i_1, i_2, i_3, i_5\}$ over $\{c_1, c_2, c_3\}$ yields ν . At P' , c_4 is again the only underdemanded school; settling i_5 there and rerunning DA among $\{i_1, i_2, i_3, i_4\}$ yields ν' . $\square$

5.2 Preferences over classmates

Duque-Rosas, Pereyra, and Torres-Martínez (2026) consider *school-lexicographic preferences over colleagues*. Students first compare matchings by the school they receive. Conditional on receiving the same school, they may care about that school’s composition, but they are indifferent between two matchings that give them both the same school and the same set of schoolmates. Let $\mathcal{D}^c$ denote the corresponding domain of students’ preference profiles over matchings, and let $P(\succeq)$ be the strict ranking of schools induced by a profile $\succeq \in \mathcal{D}^c$. A mechanism φ defined on school rankings induces a mechanism on this richer domain by

$$\bar{\varphi}(\succeq) = \varphi(P(\succeq)).$$

Duque-Rosas, Pereyra, and Torres-Martínez (2026) show that the mechanism induced by DA is stable and strategy-proof on $\mathcal{D}^c$. The intuition separates the two possible motives for a lie. A student may lie to change her school, which ordinary strategy-proofness prevents, or she may lie while keeping her school in order to change her classmates, which local non-bossiness prevents.

Corollary 3. *In Γ^0 , if φ is efficient and DA-dominating, then $\bar{\varphi}$ is not strategy-proof on $\mathcal{D}^c$. The profitable deviation can be made by student i_1 , whose school remains c_1 under both reports for every mechanism in this class.*

Proof. Choose $\succeq_{i_1} \in \mathcal{D}^c$ inducing P_{i_1} and such that, among matchings assigning i_1 to c_1 , student i_1 strictly prefers having i_2 rather than i_3 as her classmate. Complete the profile with any $\succeq_{-i_1}$ inducing P_{-i_1} . By Theorem 1 and Lemma 2, $\bar{\varphi}(\succeq) = \nu$, where i_1 shares c_1 with i_3 . If i_1 reports any $\succeq'_{i_1} \in \mathcal{D}^c$ inducing P'_{i_1} , the outcome is ν' , where she shares c_1 with i_2 . By construction, $\nu' \succ_{i_1} \nu$, so the deviation is profitable. $\square$

The same profiles also violate *local group strategy-proofness* in the sense of Duque-Rosas, Pereyra, and Torres-Martínez (2026).⁶ At P' , the coalition $\{i_1, i_2\} \subseteq \varphi_{c_1}(P')$ can have i_1 report P_{i_1} instead of P'_{i_1} . Student i_2 then moves from c_1 to her top choice c_2 , while i_1 remains at c_1 .

5.3 The three properties are jointly, but only jointly, incompatible

Each pair of the three properties in Theorem 1 is attainable in Γ^0 . DA is DA-dominating and locally non-bossy but not efficient. A fixed-order *serial dictatorship*—students choose sequentially, in an exogenous order, their most-preferred acceptable school with a remaining seat—is Pareto efficient and non-bossy, hence locally non-bossy, but need not dominate DA.⁷ With the order $(i_4, i_5, i_2, i_3, i_1)$ at profile P , serial dictatorship assigns i_4 to c_2 , i_5 to c_3 , i_2 and i_3 to c_1 , and i_1 to c_4 , whereas $\text{DA}_{i_1}(P) = c_2$. Finally, full-consent EADA is efficient and DA-dominating but locally bossy by Corollary 2. Table 2 summarizes.

	DA-dominating	Efficient	Locally non-bossy
DA	yes	no	yes
SD (order i_4, i_5, i_2, i_3, i_1)	no	yes	yes
EADA (full consent)	yes	yes	no

Table 2: Each pair of properties is attainable in Γ^0 ; the three together are not.

6 Concluding remarks

The literature on improvements over deferred acceptance has largely asked what must be given up in terms of stability, incentives, or consistency. Local non-bossiness adds a different dimension. DA allows a student’s report to affect other parts of the market, but it prevents that student from rearranging her own school while staying there. We show that this protection behaves differently across capacity structures. Under unit capacities, it is automatic for every DA-dominating mechanism. With multi-seat schools, an efficient DA improvement can be forced to give it up.

The distinction matters when students care about their peers. In our example, an efficiency-enhancing mechanism gives a student no leverage over the school she receives—she obtains the same school under both reports—but gives her leverage over who attends that school with her. Thus the welfare correction creates a strategic margin that is absent under DA. A complete

⁶Local group strategy-proofness rules out a joint misreport by a coalition whose members are initially assigned to the same school when every member weakly benefits and at least one member strictly benefits. Formally, there must not exist $x \in C \cup \{\emptyset\}$, a profile P , a coalition $K \subseteq \varphi_x(P)$, and reports P'_K such that $\varphi_i(P'_K, P_{-K}) R_i \varphi_i(P)$ for every $i \in K$ and the preference is strict for some member.

⁷Serial dictatorship is non-bossy because, if a student changes her report but chooses the same school at her turn, the vector of remaining capacities after her choice is unchanged. Hence all later students face the same choices and the entire matching is unchanged.

characterization of the priority and capacity structures for which efficient, DA-dominating, and locally non-bossy mechanisms coexist remains open.

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